How Do Treasury Rates Affect You?

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Abstract

This study examines how movements in the U.S. 10-year Treasury yield affect consumer lending rates, including mortgage, auto, personal, commercial, student loan, and credit card APRs. Using monthly data from August 2006 to April 2025, correlation and regression analyses were conducted to measure the strength of these relationships. Outlier diagnostics and assumption checks confirmed model validity, and backward elimination refined predictors to ensure statistical significance. Results revealed that secured lending products, particularly mortgages (r = .96) and commercial loans (r = .93), are susceptible to Treasury yields, while unsecured products such as credit cards (r = .47) and Stafford loans (r = .68) showed weaker or negative coefficients due to risk premiums and regulatory formulas. The final regression model explained 94% of the variance in Treasury yields, and a constant-spread benchmark confirmed the Treasury–mortgage linkage with low error rates, though spreads proved dynamic over time. Limitations include the exclusion of macroeconomic indicators and structural breaks. Findings highlight the Treasury yield's role as a macroeconomic driver of borrowing costs, offering practical value for lenders, policymakers, investors, and consumers in anticipating rate adjustments and financial market activity.

1. Introduction

1.1 Purpose of the project

The purpose of this analysis is to investigate how movements in the U.S. 10-year Treasury rate affect various consumer lending rates in the U.S. economy. The Treasury yield is often considered a risk-free benchmark rate and forms the basis for pricing other interest rates across the financial market.

1.2 Problem statement

This study focuses on the following research question: How strongly do changes in the 10-year Treasury yield predict movements in mortgage rates, commercial loan rates, student loan rates, personal loan rates, auto loan rates, and credit card APRs?

1.3 Preliminary examination

According to a preliminary analysis of financial market data, changes in the yield on the U.S. 10-year Treasury are closely correlated with changes in consumer lending rates, especially mortgage rates. For example, historically, 30-year fixed mortgage rates have followed the yield on the 10-year Treasury, typically staying one to two percentage points higher (Brevitas, 2025). By measuring the strength of these linkages, determining which lending rates are most susceptible to changes in the Treasury, and modeling predictive trends, analytics can offer insightful information. The goal of this study can be supported by such an analysis, which can clarify how benchmark rates translate into consumer borrowing costs. Subsequent correlation, regression, and modeling analyses for the team's project on interest rate behavior and financial risk modeling will be directly influenced by the results of this section.

2. Related background

Historically, the 10-year Treasury yield is a key economic indicator used by financial institutions to set rates for long-term lending products such as mortgages and personal loans. Because Treasury securities are backed by the U.S. government, their yields represent a near "risk-free" baseline. In contrast, consumer loan rates include additional risk premiums tied to borrower creditworthiness, liquidity constraints, and macroeconomic conditions.

Previous financial studies (e.g., Federal Reserve Economic Data) show that mortgage rates typically maintain a strong positive correlation with the 10-year Treasury yield, while short-term consumer credit rates (such as credit card rates) are more influenced by central bank policy and bank-level pricing strategies. (Federal Reserve Bank of Minneapolis, 2025)

3. Data

3.1 Project data description

The dataset used in this analysis consists of monthly observations of U.S. Treasury yields and consumer loan rates between August 2006 and April 2025. The yield on the 10-year Treasury, represented as an annual percentage rate, is the main independent variable. The 30-year fixed mortgage rate, credit card annual percentage rate, 5-year auto loan rate, personal loan rate, Federal Stafford graduate loan rate, and commercial 504 business loan rate are examples of dependent variables. Every observation relates to a particular date, and all rates are expressed as percentages. Because of the time series structure of the dataset, co-movements and predicted correlations between Treasury yields and consumer

lending rates can be examined. Before analysis, variables were normalized for consistency, and small missing values were filled in using interpolation.

3.2 Project data sources

Historical rate data was gathered from StatMethods.net, Federal Reserve publications, a business loan bank, and a historical federal student interest rate tracker. The references section includes specific citations. (Board of Governors of the Federal Reserve System, 2025; Freddie Mac, 2025; SBA, 2021; Kantrowitz, 2023)

3.2 Exploratory data analysis

Monthly observations of consumer lending rates and the 10-year Treasury yield from August 2006 to April 2025 are included in the dataset. Annual percentages are used to express all rates. To highlight the dataset's salient features, exploratory data analysis was carried out. For every variable, descriptive statistics were calculated. The 10-year Treasury yield ranged from 0.60% to 4.88%, with an average of 2.70% (SD = 1.04). Mortgage rates ranged from 2.68% to 7.62%, with an average of 4.72% (SD = 1.25). With an average of 14.32% (SD = 2.91), credit card APRs were generally higher but comparatively constant. Table 1 presents complete descriptive statistics. Visual analysis of time-series plots revealed that while credit card APRs stayed mostly constant, mortgage rates closely followed Treasury yields, which are found in Figure 1. Treasury yields and mortgage rates have a substantial positive link (r =.94), according to a preliminary correlation study; credit card and personal loan rates have lesser correlations. (Federal Reserve Bank of St. Louis, 2025)

3.3 Data screening

The dataset was examined for outliers and checked against the multiple regression assumptions before regression analyses were performed. No missing values were detected. Numeric data were isolated from date variables for analysis. Cook's distance, leverage values, Mahalanobis distance, and diagnostic plots were used to identify outliers and verify assumptions.

3.3.1 Outliers

Cook's distance revealed nine significant observations, Mahalanobis distance found two multivariate outliers, and leverage values highlighted five examples. Thirteen examples with extreme results were found by a combined outlier index, and these were eliminated from further analysis. There were 147 valid observations in the dataset after elimination. These are shown in Figure 2.

3.3.2 Additivity Assumption

The correlation matrix showed moderate to strong correlations between the predictors, indicating some overlap. Despite the close relationship between several variables, the model as a whole maintained enough independence to support inclusion.

3.3.3 Linearity Assumption

The dependent variable and predictors have a roughly linear relationship, according to standardized residual plots. Although there were a few small departures, the general pattern confirmed the linearity assumption.

3.3.4 Normality Assumption

A normalized residuals histogram revealed a distribution that was roughly normal with a small negative skewness. The normalcy assumption was supported by a Q-Q plot, which verified that the majority of residuals matched the predicted diagonal.

3.3.5 Homogeneity and Homoscedasticity Assumptions

Examining residual plots visually revealed that variance was largely consistent across fitted values. The homogeneity and homoscedasticity conditions were satisfied since the points were uniformly distributed above and below the zero line.

4. Technical Approach

4.1 Develop and Test Hypotheses

4.1.1 Main Hypothesis

H_0 (Null Hypothesis): There is no statistically significant relationship between the 10-year U.S. Treasury rate and consumer loan interest rates (mortgage, auto, personal, commercial, student, and credit card rates).

H_1 (Alternative Hypothesis): There is a statistically significant relationship between the 10-year U.S. Treasury rate and consumer loan interest rates.

4.1.2 Sub-Hypotheses

1. Loan Type Sensitivity: The strength of correlation with the 10-year Treasury rate varies across loan types. Long-term loans such as mortgages are expected to have stronger relationships with Treasury yields than short-term or revolving credit products like credit cards.

2. Spread Model Validity: Loan rates can be modeled as the 10-year Treasury rate plus a

fixed spread (Loan Rate = 10-Year Treasury Rate + c). However, deviations from this model over time would indicate that spreads change and other factors influence loan pricing.

3. Temporal Stability: The relationship between the 10-year Treasury rate and loan rates remains consistent over time. If the spread changes significantly across years, it suggests that the model requires periodic recalibration.

4.2 Analytics Approach Description

To test the hypothesis, a quantitative analytical approach was used in R with the following workflow:

4.1.1 Data Preparation

Imported the dataset into R and verified that all variables (dates, numeric rates) were properly written. Before conducting the correlation and regression analysis, non-numeric identifiers were removed. Standardize rate variables into uniform percentage representations.

4.1.2 Exploratory Analysis

To generate descriptive statistics, utilized psych::describe() (mean, SD, min, max, skewness, and kurtosis). Correlations between Treasury yields and loan rates were visualized using corrplot, with the 10-year Treasury yield serving as the benchmark variable. To visually analyze the correlations between the 10-year Treasury yield and each loan rate, we created scatterplots using regression lines (ggplot2).

4.1.3 Data Screening

Cook's distance, leverage values, and Mahalanobis distance were used to detect outliers. To increase the stability of the model, 13 extreme examples that were found by a combined outlier index were eliminated. Assumption checks were carried out for Additivity**,** Linearity**,** Normality, Homogeneity, and Homoscedasticity.

4.1.4 Modeling and Refinement

Created a multivariate regression model using the 10-year Treasury yield as the dependent variable and loan rates as the predictors. Results were compared before and after outlier elimination, and they showed an improvement in fit and stability. We used backward elimination (p-value technique) to iteratively remove non-significant predictors, resulting in a more compact model. To quantify specific connections and slopes, we ran simple regressions on each loan rate against the 10-year Treasury yield. The reported coefficients, significance levels, and model fit statistics are in APA format. Together, the regression refinement and constant-spread benchmark provided complementary perspectives, setting the stage for evaluating model robustness and predictive validity in the next section.

4.1.5 Interpretation and Limitations

The findings demonstrated that whereas unsecured lending products (credit cards, personal loans) are less responsive because of risk premiums and consumer credit criteria, secured lending products (mortgages, auto loans, commercial loans) closely follow changes in the 10-year Treasury yield. Potential structural changes in lending markets and the omission of other macroeconomic factors (such as inflation expectations and the Federal Funds Rate) are among the limitations. These interpretations, including

the strengths and weaknesses of the constant-spread model, guided the subsequent evaluation of model performance and informed the discussion of broader implications.

4.1.6 Tools and Reproducibility

All analyses were carried out in RStudio (v4.4) using the readxl, corrplot, ggplot2, dplyr, car, lmtest, and psych packages. The outputs were generated using R Markdown and exported to Word using APA formatting.

5. Test and Evaluation

5.1 Methodology

The regression [and constant spread] model's validity was assessed using outlier diagnostics, assumption checks, and model refining. Outliers were discovered using Mahalanobis distance, leverage values, and Cook's distance, leading to the removal of 13 cases. Correlation matrices and diagnostic plots were used to evaluate the assumptions of additivity, linearity, normalcy, and homoscedasticity, and all of them were satisfied. Model robustness was verified by comparing results before and after outlier removal, and backward elimination was used to ensure that only meaningful predictors remained. These stages guaranteed that the final model was both statistically valid and easily interpretable.

5.2 Correlation

The 10-year Treasury rate shows strong positive correlations with mortgage (0.96), commercial business loan (0.93), 5-year auto (0.85), and personal loan (0.87) rates. Interestingly, the 10-year Treasury yield rates had the weakest correlation with the CC rates at 0.47. Additionally, the Federal Stafford graduate rates had lower correlations with

all the treasury yields, with the 10-year only holding a strength at 0.68. The Full correlation table is displayed in Table 2. (Federal Reserve Bank of St. Louis, 2025)

5.3 Regression

5.3.1 Model Predictors

5.3.1.1 Intercept

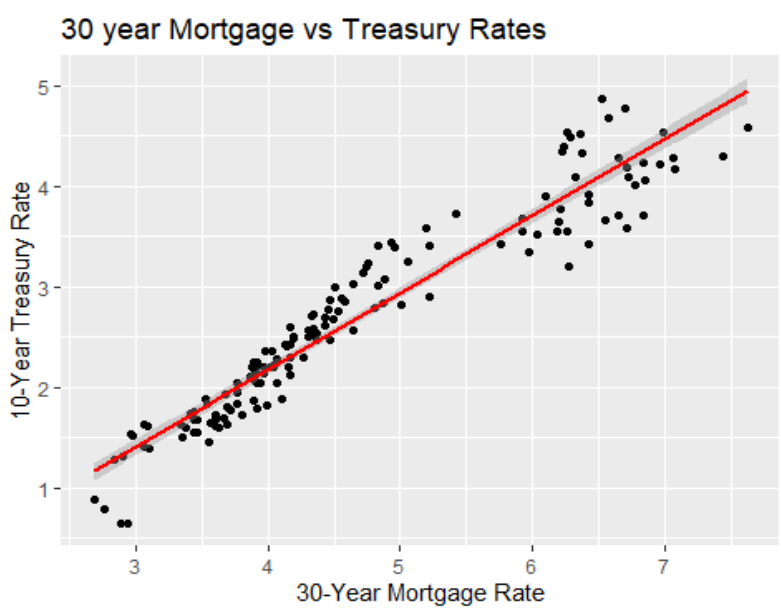
$X_0 = -1.025$

–1.025 is the baseline value of the Treasury rate when all predictors are zero, which is not practically meaningful, but part of the model.

5.3.1.2 30-year Mortgage

X_1 = 30-year Mtg Rate: $\beta_1 = 0.588$, $R^2 = 0.913$, p-value = 1.39e-07

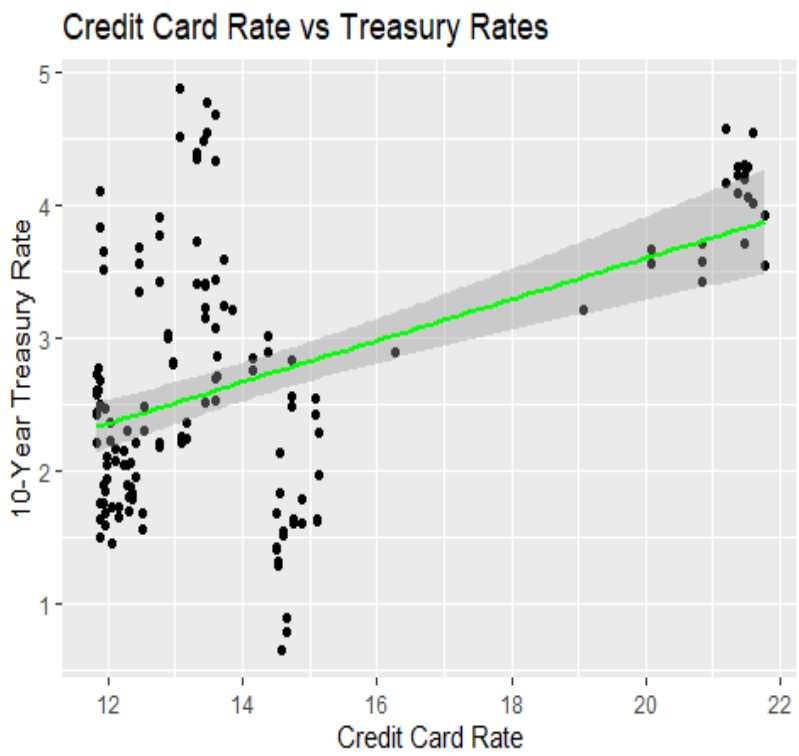
For every 1 percentage point increase in a 30-year mortgage rate, the 10-year Treasury rate is expected to rise by 0.588 points, holding other lending rates constant, showing a strong positive correlation.



5.3.1.3 Credit Card Rate

X_2 = Credit Card Rate: β_2 = - 0.044, R^2 = 0.213, p-value = 0.001

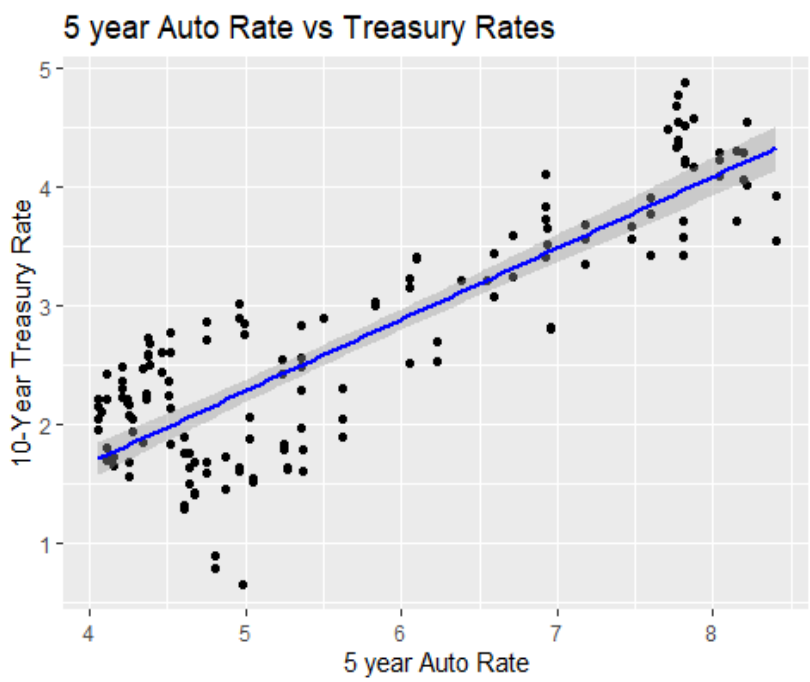
A 1-point increase in average credit card rates is associated with a 0.044-point decrease in the Treasury rate. Weak and negative, which suggests CC rates don't track Treasuries closely, likely because they're driven more by risk premiums and consumer credit factors from banks, credit unions, or fintech lenders.



5.3.1.4 5-year car loan

X_3 = 5-year car loan rate: β_3 = 0.075, R^2 = 0.725, p-value = 0.160

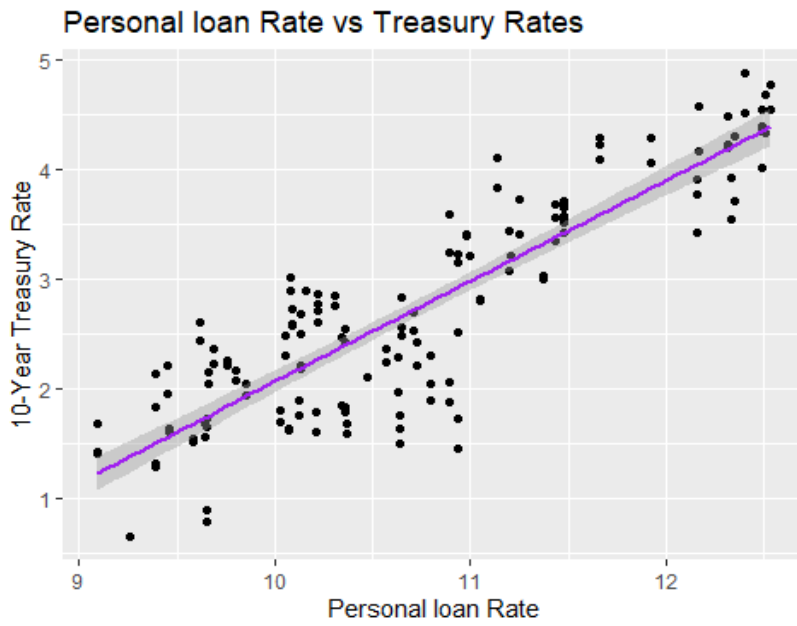
A 1-point increase in auto loan rates corresponds to a 0.075-point increase in the Treasury rate. There is a positive but moderate correlation since these move with broader interest rate trends, but the strongest drivers are credit risk, loan term, and vehicle type.



5.3.1.5 Personal loan

X_4 = Personal loan rate: $\beta_4 = 0.052$, $R^2 = 0.764$, p-value = 0.442

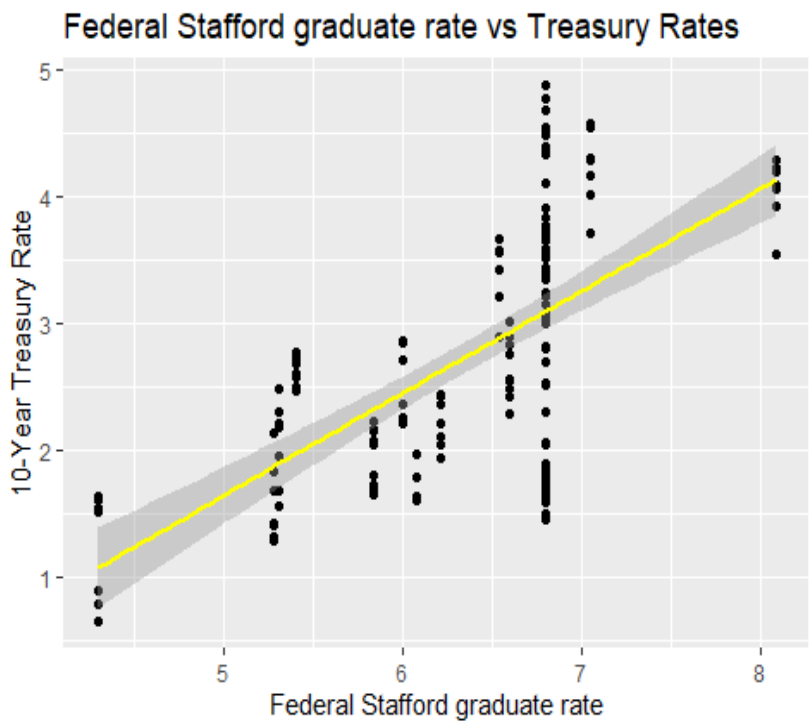
A 1-point increase in personal loan rates corresponds to a 0.052-point increase in the Treasury rate. There is a small effect since personal loan pricing is less sensitive to Treasury movements.



5.3.1.6 Federal Stafford graduate loan

X_5 = Federal Stafford graduate loan rate: β_5 = - 0.052, R^2 = 0.455, p-value = 0.231

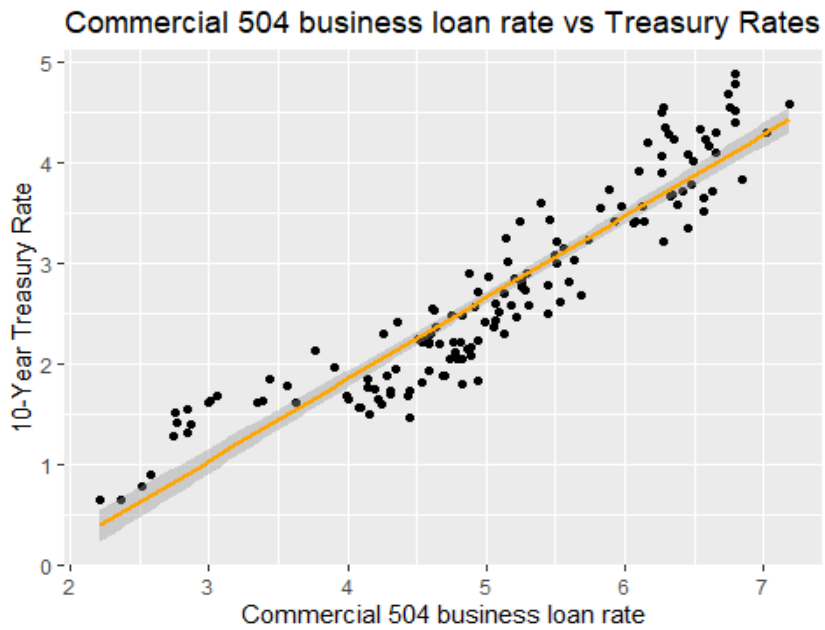
A 1-point increase in Stafford graduate loan rates is associated with a 0.052-point decrease in the Treasury rate. This negative sign is counterintuitive, since Stafford rates are formulaically tied to Treasuries. This may reflect timing differences (rates reset annually) or multicollinearity in the model.



5.3.1.5 Commercial 504 business loan

X_6 = Commercial 504 business loan rate: $\beta_6 = 0.184$, $R^2 = 0.863$, p-value = 0.033

Commercial 504 business loan rate: A 1-point increase in SBA 504 commercial loan rates corresponds to a 0.184-point increase in the Treasury rate. Strong positive correlation — business lending tracks Treasury yields closely.



5.3.2 Overall Best Model

As shown on figure 3 and below, overall, the model is significant with a p-value < 2.2e-16. Our final model had a $R^2 = 0.938 \approx .94$, indicating that the loan rate variables we included explained almost 94% of the variation in the 10-year Treasury yield. Although the fit is excellent, it is crucial to note that R^2 demonstrates explanatory power rather than causality. Having a high R^2 does not ensure meeting assumptions (which is why we performed diagnostic testing). It also does not imply that every predictor is significant; p-values and effect sizes are used to determine this, which is why, by utilizing the Backward elimination using p-value, we will find the most significant consumer rate predictors. We will need to find the best model with all predictors significant (presuming a 0.05 significance level) since we want to take precautions against model overfitting.

	Best model with all variable predictors significant					
Step	30-year Mtg Rate	Credit Card Rate	5-year car loan rate	Personal loan rate	Federal Stafford graduate rate	Commercia l 504 business loan rate
Step 1 - Full model	1.39e-07	0.0014	0.1600	*0.4418*	0.2312	0.0330
Step 2 – Personal loan rate removed	1.59e-07	0.0009	0.0202		*0.3116*	0.0177
Step 3 – Federal Stafford graduate rate removed	1.71e-09	0.0001	0.0359			0.0303

5.3.3 Best Model Equation

y = -1.184 + 0.588(30-year Mtg Rate) - 0.044(Credit Card Rate) + 0.075(5-year car loan rate) + 0.184(Commercial 504 business loan rate)

5.3.4 Overall Best Model Observations

Due to their dependence on Treasury benchmarks for long-term loan pricing, mortgage and commercial loan rates turned out to be the best indicators of Treasury yields. Because of risk premiums, regulations, and yearly reset formulae that lessen sensitivity to changes in the Treasury, credit card, and Stafford loan rates, they displayed weak or negative coefficients. In addition to borrower-specific risk indicators, auto and personal loans showed moderately positive correlations, suggesting a partial dependency on Treasury yields. The Treasury's function as a macroeconomic benchmark was highlighted by the regression model's overall explanation of 94% of the variance in Treasury rates, which also confirmed that loan type sensitivity differs across products.

5.4 Mortgage Constant-Spread Prediction Model

To assess predictive validity, a constant-spread model was applied to mortgage rates, shown in Figures 4 and 5. The model assumes mortgage rates are equal to the 10-year Treasury yield plus a fixed spread, calibrated at 1.782 percentage points using the August-December 2006 baseline. Predictions for the entire sample period were compared to actual mortgage rates, with errors measured using RMSE and MAE. The model achieved RMSE = 0.033 and MAE = 0.031, suggesting high predictive accuracy, as shown in Figure 4. The constant-spread technique captured the broad relationship, although variations occurred during periods of market volatility, indicating that spreads are dynamic and not fixed. This analysis validates the Treasury yield's role as a benchmark for mortgage pricing and emphasizes the need to include time-varying risk premiums in future models.

6. Discussion

6.1 Interpretation of findings

The study found that secured lending products, such as mortgages and commercial loans, are more susceptible to changes in the 10-year Treasury yield than unsecured credit products, such as credit cards. Regression analysis indicated that the model accounted for 94% of the variation in Treasury yields, with mortgage and commercial loan rates appearing as the most powerful predictors. The constant-spread model verified the Treasury-mortgage linkage, with low error rates. However, variations during turbulent periods show the dynamic character of spreads. Overall, the findings support the

rejection of the null hypothesis and demonstrate that Treasury yields serve as a benchmark for consumer loan pricing, with sensitivity changing by loan type.

6.2 Business implications

The findings have strong business consequences. For financial institutions, Treasury yields are a dependable benchmark for pricing secured lending products, particularly mortgages and commercial loans. Policymakers can see how changes in Treasury yields affect consumer borrowing costs, highlighting the Treasury's involvement in monetary transmission. Consumers benefit from understanding that mortgage and vehicle loan rates are closely related to Treasury yields, whereas credit card rates are less predictable due to risk premiums. Finally, Treasury movements can serve as an early predictor of housing and credit market activity. These consequences illustrate the Treasury yield's significant impact on financial decision-making across sectors. (Federal Reserve Bank of Minneapolis, 2025)

6.3 Limitations of the study

There are various limitations to this study. First, the dataset was limited to monthly observations of particular loan kinds, which would not have captured larger lending categories or short-term volatility. Second, financial relationships are frequently nonlinear and susceptible to structural breakage, even though regression assumptions were evaluated and found to be met. Although spreads fluctuate over time due to credit risk and liquidity constraints, the constant-spread model assumes fixed mortgage spreads. (Jackson, 2025) Third, certain variables, especially for Stafford loans, were non-significant, indicating either multicollinearity or formulaic resets. Lastly, macroeconomic factors like unemployment and inflation, which probably affect both

consumer loan rates and Treasury yields, were not included in the analysis. Despite the statistical validity of the results, these limitations imply that they should be considered as correlational rather than causal benchmarks.

6.4 Suggestions for future research

In order to capture more comprehensive drivers of loan pricing, future research could extend the model by including macroeconomic indicators like inflation, unemployment, and the Federal Funds Rate. Time-series techniques such as ARIMA or VAR could be used to take into consideration structural breaks and lagged effects, especially in times of financial strain. By modeling nonlinear interactions, machine learning techniques may also increase prediction accuracy. (Wang, Wang, & Tu, 2025) Furthermore, a dynamic framework where spreads change in response to risk and liquidity conditions might be developed from the constant-spread model. Lastly, cross-market comparisons can investigate if Treasury yields have a comparable impact on loan rates in other economies. These paths would improve the findings' resilience and applicability. (Jackson, 2025)

7. Conclusion and Future Work

7.1 Concise summary of key insights

The study indicated that the 10-year Treasury yield is a major predictor of consumer lending rates, especially for secured items like mortgages and commercial loans. Correlation and regression analysis revealed robust connections, with the final model accounting for 94% of the variation in Treasury yields. A constant-spread model supported the Treasury-mortgage relationship, although spreads fluctuated over time. The null hypothesis was denied, but sub-hypotheses were validated, indicating that loan

type sensitivity varies and Treasury yields are still a reliable benchmark. These findings highlight the Treasury yield's position as a macroeconomic driver of borrowing rates, with important consequences for lenders, policymakers, and consumers.

7.2 Practical recommendations

These findings lead to useful recommendations. Treasury yields should be constantly watched by lenders as standards for secured lending products, especially commercial loans and mortgages. While revolving credit is still less sensitive, policymakers are aware that changes in the Treasury immediately affect the cost of borrowing for consumers. Customers may expect mortgage and auto loan rates to change in tandem with Treasury yields, which will influence borrowing and refinancing choices. Lastly, changes in Treasury yields can be used by investors to gauge activity in the housing and credit markets. When taken as a whole, these suggestions emphasize how crucial the Treasury yield is in determining financial strategies for many industries. (Dickler, 2025)

7.3 Future Work

Future studies might include macroeconomic data like inflation and unemployment, as well as time-series approaches like ARIMA and VAR, to account for lagged impacts and structural breaks. Modeling nonlinear interactions using machine learning techniques like Random Forest Regression can increase predicted accuracy. The constant-spread paradigm can be transformed into a dynamic framework where spreads vary based on risk and liquidity levels. Cross-market comparisons can identify global benchmark impacts, while borrower segmentation can highlight sensitivity differences between credit

risk groups. Finally, forecasting applications may convert these models into useful tools for lenders and policymakers. (Wang, Wang, & Tu, 2025)

7.4 Business Value

Gaining insight into the relationship between consumer loan rates and Treasury yields offers real commercial benefits. It makes proactive loan pricing and risk management possible for financial organizations. Policymakers learn how monetary policy affects the cost of borrowing for consumers. (Dickler, 2025) While consumers gain from increased openness in predicting borrowing prices, investors can use Treasury changes as leading indicators of housing and credit market activity. When taken as a whole, these observations demonstrate how crucial the Treasury yield is in influencing financial strategies in a variety of industries.

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Appendices:

Table 1. Descriptive statistics

	vars	n	mean	sd	median	trimmed	mad	min	max
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
observation_date*	1	161	81.00	46.62	81.00	81.00	59.30	1.00	161.00
Oneyr_Tsy_Rate	2	161	1.65	1.80	0.65	1.42	0.82	0.00	5.44
Fiveyr_Tsy_Rate	3	161	2.21	1.29	1.84	2.15	1.26	0.00	4.86
Tenyr_Tsy_Rate	4	161	2.70	1.04	2.56	2.68	1.24	0.60	4.88
Thirtyyr_Mtg_Rate	5	161	4.72	1.25	4.35	4.65	1.11	2.68	7.62
CC_Rate	6	161	14.32	2.91	13.44	13.74	1.79	11.82	21.76
Fiveyr_auto_loan_rate	7	161	5.72	1.38	5.25	5.63	1.45	4.05	8.40
Personal_loan_rate	8	161	10.66	0.97	10.63	10.62	1.10	8.73	12.53
Federal_Stafford_graduate_rate	9	161	6.31	0.83	6.60	6.35	0.58	4.30	8.08
Commercial_504_business_loan_rate	10	161	5.05	1.19	5.06	5.13	1.19	2.21	7.18

Figure 1. Monthly comparison of interest for various loan types and the 10-year treasury

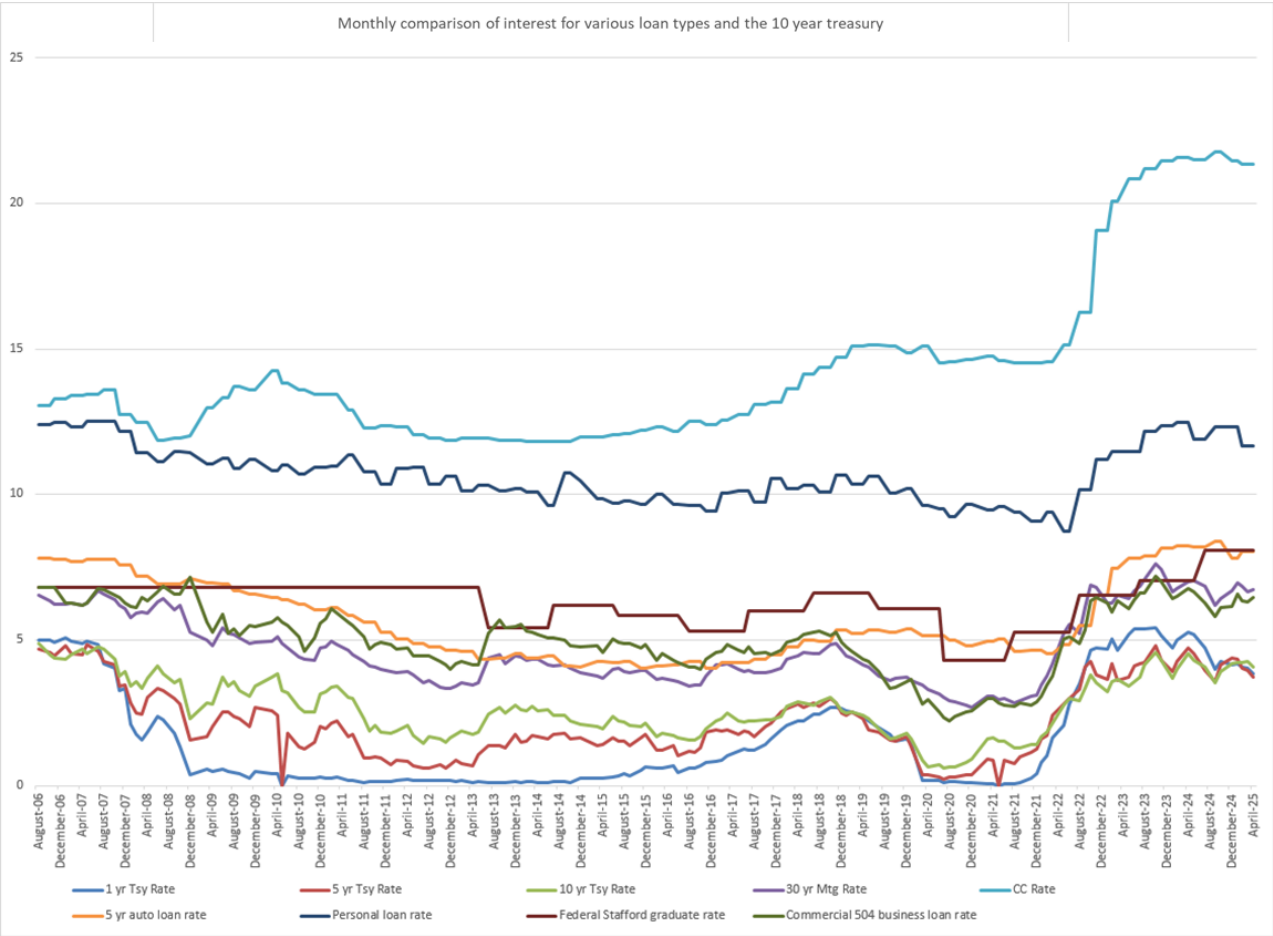

Figure 2. Outlier tests

Mahal outlier test

This found 2 outliers in the dataset

```
#Mahalanobis outlier test:

mahal_test <- rates_project[,-1]
mahal_check <- mahalanobis(mahal_test,
                    center = colMeans(mahal_test),
                    cov = cov(mahal_test))
cutoff_mahal <- qchisq(1-.001, ncol(mahal_test))
mahal_outliersareproblem <- as.numeric(mahal_check > cutoff_mahal)
table(mahal_outliersareproblem)

## mahal_outliersareproblem
##   0   1
## 159   2
```

Leverage outlier test

This found 5 outliers in the dataset

```
leverage_test <- rates_project[,-1]
initial_model <- lm(Tenyr_Tsy_Rate ~ Thirtyyr_Mtg_Rate
 + CC_Rate
 + Fiveyr_auto_loan_rate
 + Personal_loan_rate
 + Federal_Stafford_graduate_rate
 + Commercial_504_business_loan_rate,
             data = leverage_test)


k <- 6 ##number of IVs
leverage_check <- hatvalues(initial_model)
cut_leverage <- (2*k+2) / nrow(leverage_test)
bad_leverage <- as.numeric(leverage_check > cut_leverage)
table(bad_leverage)

## bad_leverage
##   0   1
## 156   5
```

Cook's outlier test:

This found 9 outliers in the dataset

```
cooks_test <- rates_project[,-1]

cooks_check <- cooks.distance(initial_model)
cut_the_cooks <- 4 / (nrow(cooks_test) - k - 1)
outlier_cooks <- as.numeric(cooks_check > cut_the_cooks)
table(outlier_cooks)

## outlier_cooks
##   0   1
## 152   9
```

Outliers combined

```
total_outliers <- mahal_outliersareproblem + bad_leverage + outlier_cooks
table(total_outliers)

## total_outliers
##   0   1   3
## 147  13   1
```

Remove outliers:

13 outliers have been removed

```
outlier_chk <- rates_project[,-1]

rem_outliers <- subset(outlier_chk, total_outliers < 1)

nrow(rem_outliers)

## [1] 147
```

Table 2. Correlation matrix

```
##                                   Tenyr_Tsy_Rate Thirtyyr_Mtg_Rate   CC_Rate
## Tenyr_Tsy_Rate                         1.0000000         0.9559189 0.4673221
## Thirtyyr_Mtg_Rate                      0.9559189         1.0000000 0.6153824
## CC_Rate                                0.4673221         0.6153824 1.0000000
## Fiveyr_auto_loan_rate                  0.8523824         0.8996545 0.6514161
## Personal_loan_rate                     0.8747902         0.8832560 0.4755138
## Federal_Stafford_graduate_rate         0.6772280         0.6974662 0.4150975
## Commercial_504_business_loan_rate      0.9294240         0.9180723 0.3405642
##                                   Fiveyr_auto_loan_rate Personal_loan_rate
## Tenyr_Tsy_Rate                                0.8523824          0.8747902
## Thirtyyr_Mtg_Rate                             0.8996545          0.8832560
## CC_Rate                                       0.6514161          0.4755138
## Fiveyr_auto_loan_rate                         1.0000000          0.9007661
## Personal_loan_rate                            0.9007661          1.0000000
## Federal_Stafford_graduate_rate                0.6911013          0.7768259
## Commercial_504_business_loan_rate             0.7425711          0.8408836
##                                   Federal_Stafford_graduate_rate
## Tenyr_Tsy_Rate                                          0.6772280
## Thirtyyr_Mtg_Rate                                       0.6974662
## CC_Rate                                                 0.4150975
## Fiveyr_auto_loan_rate                                   0.6911013
## Personal_loan_rate                                      0.7768259
## Federal_Stafford_graduate_rate                          1.0000000
## Commercial_504_business_loan_rate                       0.7167083
##                                   Commercial_504_business_loan_rate
## Tenyr_Tsy_Rate                                            0.9294240
## Thirtyyr_Mtg_Rate                                         0.9180723
## CC_Rate                                                   0.3405642
## Fiveyr_auto_loan_rate                                     0.7425711
## Personal_loan_rate                                        0.8408836
## Federal_Stafford_graduate_rate                            0.7167083
## Commercial_504_business_loan_rate                         1.0000000
```

Figure 3. Regression models

```
second_model <- lm(Tenyr_Tsy_Rate ~ Thirtyyr_Mtg_Rate
                   + CC_Rate
                   + Fiveyr_auto_loan_rate
                   + Personal_loan_rate
                   + Federal_Stafford_graduate_rate
                   + Commercial_504_business_loan_rate,
                   data = rem_outliers)

summary(second_model)

##
## Call:
## lm(formula = Tenyr_Tsy_Rate ~ Thirtyyr_Mtg_Rate + CC_Rate + Fiveyr_auto_loan_rate +
##     Personal_loan_rate + Federal_Stafford_graduate_rate + Commercial_504_business_loan_rate,
##     data = rem_outliers)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.57452 -0.14899  0.00416  0.19492  0.49771
##
## Coefficients:
##                                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)                       -1.02512    0.46475  -2.206  0.02903 *
## Thirtyyr_Mtg_Rate                  0.58823    0.10601   5.549 1.39e-07 ***
## CC_Rate                           -0.04353    0.01338  -3.253  0.00143 **
## Fiveyr_auto_loan_rate              0.07493    0.05305   1.412  0.16006
## Personal_loan_rate                 0.05225    0.06775   0.771  0.44185
## Federal_Stafford_graduate_rate    -0.05163    0.04293  -1.202  0.23121
## Commercial_504_business_loan_rate  0.18413    0.08551   2.153  0.03302 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2487 on 140 degrees of freedom
## Multiple R-squared:  0.9408, Adjusted R-squared:  0.9382
## F-statistic: 370.6 on 6 and 140 DF,  p-value: < 2.2e-16
```

```r
third_model <- lm(Tenyr_Tsy_Rate ~ Thirtyyr_Mtg_Rate
                  + CC_Rate
                  + Fiveyr_auto_loan_rate
                  + Federal_Stafford_graduate_rate
                  + Commercial_504_business_loan_rate,
                  data = rem_outliers)

summary(third_model)

##
## Call:
## lm(formula = Tenyr_Tsy_Rate ~ Thirtyyr_Mtg_Rate + CC_Rate + Fiveyr_auto_loan_rate +
##     Federal_Stafford_graduate_rate + Commercial_504_business_loan_rate,
##     data = rem_outliers)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.59461 -0.15146  0.01178  0.18455  0.51119
##
## Coefficients:
##                                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)                       -0.70230    0.20168  -3.482 0.000663 ***
## Thirtyyr_Mtg_Rate                  0.58290    0.10563   5.518 1.59e-07 ***
## CC_Rate                           -0.04500    0.01323  -3.403 0.000869 ***
## Fiveyr_auto_loan_rate              0.09951    0.04236   2.349 0.020199 *
## Federal_Stafford_graduate_rate    -0.04141    0.04078  -1.015 0.311622
## Commercial_504_business_loan_rate  0.19941    0.08307   2.400 0.017678 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2484 on 141 degrees of freedom
## Multiple R-squared:  0.9405, Adjusted R-squared:  0.9384
## F-statistic: 445.9 on 5 and 141 DF,  p-value: < 2.2e-16
```

```r
fourth_model <- lm(Tenyr_Tsy_Rate ~ Thirtyyr_Mtg_Rate
                   + CC_Rate
                   + Fiveyr_auto_loan_rate
                   + Commercial_504_business_loan_rate,
                   data = rem_outliers)

summary(fourth_model)

##
## Call:
## lm(formula = Tenyr_Tsy_Rate ~ Thirtyyr_Mtg_Rate + CC_Rate + Fiveyr_auto_loan_rate +
##     Commercial_504_business_loan_rate, data = rem_outliers)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.5887 -0.1568  0.0037  0.1835  0.5228
##
## Coefficients:
##                                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)                       -0.77700    0.18781  -4.137 5.99e-05 ***
## Thirtyyr_Mtg_Rate                  0.62525    0.09706   6.442 1.71e-09 ***
## CC_Rate                           -0.04928    0.01254  -3.929 0.000133 ***
## Fiveyr_auto_loan_rate              0.08076    0.03812   2.118 0.035896 *
## Commercial_504_business_loan_rate  0.15641    0.07148   2.188 0.030285 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2484 on 142 degrees of freedom
## Multiple R-squared:  0.9401, Adjusted R-squared:  0.9384
## F-statistic:   557 on 4 and 142 DF,  p-value: < 2.2e-16
```

Figure 4. Constant-spread model plot

Figure 5. Constant-spread model code

Compute avg_2006_spread (Aug–Dec 2006)

```
cal_start <- as.Date(params$cal_start)
cal_end   <- as.Date(params$cal_end)

df_cal <- df |> filter(date >= cal_start, date <= cal_end)
stopifnot(nrow(df_cal) > 0)

df_cal <- df_cal |> mutate(spread = mortgage - tsy10)
avg_2006_spread <- mean(df_cal$spread, na.rm = TRUE)

kable(data.frame(
  Calibration_Start = cal_start,
  Calibration_End   = cal_end,
  Avg_2006_Spread_pp = round(avg_2006_spread, 3)
), caption = "Calibration result (Aug-Dec 2006)")
```

Apply constant-spread model

```
df <- df |>
  mutate(
    mortgage_hat = tsy10 + avg_2006_spread,
    error_pp = mortgage - mortgage_hat
  )

# Simple error metrics for the whole sample
rmse <- sqrt(mean(df$error_pp^2, na.rm = TRUE))
mae  <- mean(abs(df$error_pp), na.rm = TRUE)

kable(data.frame(
  Metric = c("RMSE_pp", "MAE_pp"),
  Value  = round(c(rmse, mae), 3)
), caption = "Model error (entire sample)")
```

Plot: Actual vs Fitted (constant spread)

```
ggplot(df, aes(x = date)) +
  geom_line(aes(y = mortgage)) +
  geom_line(aes(y = mortgage_hat), linetype = "dashed") +
  labs(
    title = "Mortgage = 10Y + avg_2006_spread (Constant-Spread Model)",
    subtitle = sprintf("avg_2006_spread = %.3f pp (Aug–Dec 2006)", avg_2006_spread),
    x = NULL, y = "Rate (percentage points)"
  ) +
  theme_minimal()
```